ChangeFine: Fine-Grained Change Detection From Remote-Sensing Imagery Based on Contrastive Prompt Learning With SAM-CLIP

Kai Deng, Siyuan Wei, Xiangyun Hu, Lizhen Lei, Yibing Xiong, and Aokun Liang

***Abstract*—Change detection (CD) is crucial for remote-sensing applications, widely used in urban planning, environmental monitoring, and disaster response. However, existing CD methods face several challenges, such as difficulty in distinguishing suspected change features, ambiguous boundaries, and inconsistencies in multiscale features. To address these issues, we propose a binary CD method called ChangeFine. This method effectively fuses the detailed features extracted by the segment anything model (SAM) and the semantic priors provided by contrastive languageâ xxx sts image pre-training (CLIP), combined with adaptive mixture-of-experts fusion and change-prompt guidance, achieving precise and fine-grained change localization in bi-temporal remote-sensing imagery. In particular, we design the SAM-CLIP feature encoder (SCFE), which bridges the modality gap between structural perception and semantic understanding. We introduce the cross-perception feature module (CPFM), which integrates gated mixture-of-experts and channel–spatial attention mechanisms, enabling fine-grained interaction and adaptive fusion between bi-temporal features and effectively enhancing the extraction of change-specific features. To handle ambiguous regions (i.e., hard pixels), we introduce the hard sample contrast module (HSCM), which dynamically identifies highly uncertain samples and helps the model accurately learn discriminative boundaries. Furthermore, inspired by the concept of semantic evolution, we design the prompt change decoder (PCD), which progressively integrates explicit predictions with implicit semantic differences to refine the reconstruction of change regions, achieving a coarse-to-fine change localization. Extensive experiments conducted on four public remote-sensing CD datasets demonstrate that ChangeFine effectively enhances change representation and maintains competitive accuracy and robustness across datasets. The code is available at https://github.com/whudk/ChangeFine**

***Index Terms*—Change detection (CD), contrast learning, multimodal feature fusion, prompt learning.**

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I. Introduction

REMOTE sensing change detection (RSCD) uses high-resolution satellite imagery from the same area at different times to automatically identify and analyze dynamic changes in surface objects and environmental features, representing a critical and challenging research area within remote sensing [1]. Recently, RSCD has been widely applied in urban development, ecological conservation, resource monitoring, and disaster response, demonstrating significant practical value and societal impact.

Traditional change-detection (CD) methods can be broadly classified into pixel-based approaches (PBCD) and object-based approaches (OBCD) [1], [2]. PBCD compares pixel values directly across bi-temporal images, but is highly sensitive to noise and lacks contextual awareness, often resulting in low accuracy [3], [4]. To address these limitations, OBCD methods segment images into meaningful regions and analyze changes at the object level [5], [6]. Although OBCD improves robustness by incorporating spatial and spectral features, it still suffers from segmentation uncertainty, dependence on hand-crafted features, and sensitivity to image quality [7], [8]. With the rapid development of deep learning, convolutional neural networks (CNNs) have been widely applied to RSCD, significantly improving detection accuracy and automation [9], [10], [11]. Early CNN-based methods focused on extracting deep spatial features, while later works introduced multilevel fusion and semantic consistency constraints for better change representation. More recently, transformer-based models [12], [13], [14], [15], [16], [17] have shown great potential by modeling global context and long-range dependencies. However, despite their impressive performance on specific datasets, these methods often exhibit limited generalization when applied to diverse scenes with complex semantics, varying resolutions, or subtle changes.

Recently, foundation models such as the segment anything model (SAM) [18] and contrastive language-image pre-training (CLIP), [19] have shown remarkable generalization in various vision tasks, thanks to large-scale pretraining in massive image–text pairs or dense segmentation annotations. These models offer rich semantic priors and strong cross-domain transferability, making them promising candidates for remote-sensing applications, including CD. Some recent studies have attempted to adapt foundation models to this domain.

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For example, ChangeCLIP [20] explores the alignment of the vision language for semantic-level CD, while SAM-CD [21], [22] leverages the strong boundary localization ability of SAM for guided mask refinement.

However, the integration of foundation models into RSCD presents significant challenges. We acknowledge that most existing approaches employ these models in a plug-and-play fashion primarily due to their prohibitive computational complexity, which makes full end-to-end joint optimization difficult on standard hardware. Consequently, the absence of joint representation learning in prior works is often a result of practical computational constraints, rather than conceptual neglect. Nevertheless, distinct from prior works, we argue that despite these costs, a unified framework that jointly exploits CLIP and SAM is conceptually necessary for fine-grained CD. Crucially, CLIP and SAM possess inherently complementary limitations that cannot be resolved when used in isolation: CLIP excels at semantic abstraction but suffers from coarse spatial localization [23], whereas SAM provides precise boundary delineation but lacks semantic awareness, treating all objects as class-agnostic masks [24], [25]. Simply cascading these models leads to a spatial–semantic misalignment, where semantic changes identified by CLIP may not align with the physical boundaries extracted by SAM. Therefore, deep interaction between these two modalities is essential to achieve semantically accurate and spatially precise CD. Furthermore, with the recent advent of parameter-efficient fine-tuning (PEFT) techniques [26], [27], it has become feasible to achieve such deep integration. By keeping the massive backbones frozen and optimizing only lightweight interaction modules, we can effectively bridge the spatial–semantic gap with manageable computational overhead.

To this end, we propose ChangeFine, a prompt-guided supervised CD framework that fully exploits the complementary strengths of SAM and CLIP for fine-grained bi-temporal remote-sensing analysis. Our framework integrates structural representations from SAM, semantic priors from CLIP, and change-aware prompts to produce more precise and robust change localization. In particular, we design an SAM-CLIP feature encoder (SCFE) to bridge the modality gap between structural perception and semantic understanding, generating joint spatial–semantic representations. In addition, we introduce a cross-perception feature module (CPFM) to facilitate fine-grained interaction between bi-temporal features, where local alignment and adaptive fusion are achieved through a gated mixture-of-experts approach and a channel–spatial attention refinement mechanism. To further enhance the model's ability to discriminate subtle or ambiguous changes, we introduce a hard sample contrast module (HSCM) that identifies highly uncertain pixels and encourages better boundary learning. In addition, inspired by the concept of semantic evolution, we develop a prompt change detector (PCD) that reconstructs change regions in a coarse-to-fine manner by incorporating both explicit predictions and implicit semantic differences as prompts. As illustrated in Fig. 1, extensive experiments conducted on four publicly available CD datasets reveal that our method exhibits improved performance in both accuracy and generalization compared to existing methodologies.

The main contributions of this article are summarized as follows.

1) We propose an SCFE that effectively integrates structural representations from SAM and global semantic priors from CLIP. Using the complementary strengths of spatial precision and semantic richness, SCFE enables robust and discriminative multimodal feature representations for CD.
2) We introduce a CPFM that performs fine-grained interaction and adaptive fusion between bi-temporal features. By combining local spatial alignment and semantic enhancement through gated Mixture of Experts (MoE) and channel–spatial attention, CPFM enables the extraction of highly discriminative change-specific features, significantly improving the model's sensitivity to subtle and ambiguous changes.
3) We develop a prompt-guided decoding framework tailored for fine-grained CD, which integrates hard-sample contrastive learning and a change guide decoder. This design enhances boundary discrimination by focusing on uncertain regions and progressively refines change localization through implicit semantic cues.
4) We conducted extensive experiments on four public RSCD datasets, demonstrating that our method achieves competitive performance in terms of accuracy and robustness, highlighting its strong generalization ability and practical applicability.

II. Related Work

A. Binary CD Methods

Bi-temporal CD aims to identify surface changes between two remote-sensing images acquired at different time periods [1]. With the success of deep CNNs (DCNNs) in visual recognition tasks [28], a variety of CNN-based methods have been developed for RSCD. Early methods employed Siamese networks to independently extract features from bi-temporal images, followed by change inference using distance metrics or shallow classifiers [29]. An end-to-end fully convolutional network was introduced to fuse bi-temporal features via early fusion, concatenation, and difference operations [9]. More recently, a 3-D convolutional encoder–decoder with adjacent-level feature fusion was proposed to improve spatiotemporal feature representation [30].

To improve feature discrimination, attention mechanisms have been incorporated into CNN-based models. Global coattention was used to model cross-temporal dependencies [31], while self-attention modules were introduced for refined spatial localization [11]. A bi-temporal transformer was also proposed to capture spatial–temporal correlations [32]. In addition, channel and spatial attention mechanisms (CBAM) were adopted to enhance feature refinement [33], [34], [35].

More recently, pure transformer-based architectures have been explored for long-range contextual modeling. A model based on the MiT backbone with overlapping patch embeddings was proposed to enhance spatial interactions across bi-temporal features [13]. The shifted window strategy from Swin-T was adopted to improve intrapatch information flow [36], [37], while global dependency modeling was achieved

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using the ViT backbone in hyperspectral CD [38], [39]. In addition to transformers, recent methods have begun to explore state space models (SSMs) [40] for CD. ChangeMamba [41] introduces a gated SSM-based architecture that captures long-range dependencies with lower computational cost and improved adaptability in complex bi-temporal scenarios. However, despite these advancements, most existing methods still rely on data-specific encoders and lack strong semantic understanding, resulting in limited generalization across heterogeneous remote-sensing datasets. To address this, some recent works have attempted to incorporate foundation models into the CD pipeline. For example, ChangeCLIP [20] explores CLIP's vision language alignment to introduce semantic priors, while SAM-CD [22] uses the mask decoder of SAM to refine spatial localization. These approaches demonstrate the potential of foundation models to provide richer and more transferable prior knowledge, but are often used in a plug-and-play manner without deep integration or task-aware adaptation.

Despite recent progress, existing foundation-model-based CD methods are still largely limited to plug-and-play use, with insufficient task-aware adaptation. As a result, the complementary advantages of CLIP in semantic understanding and SAM in structural localization are not fully exploited. To address this issue, we propose a unified framework that combines CLIP and SAM through PEFT. By freezing the heavy backbones and introducing lightweight interaction modules, our method achieves deep spatial–semantic fusion, improving both semantic consistency and boundary precision with modest computational overhead.

B. Coarse-to-Fine CD With Contrastive Learning

The coarse-to-fine CD paradigm improves accuracy via a hierarchical refinement mechanism that initiates with low-resolution global predictions to discern macroscopic patterns, subsequently integrating fine-grained features to achieve precise localization of subtle changes and accurate boundary delineation [42], [43], [44]. Building upon this multiscale framework, contrastive learning further strengthens feature discriminability by dynamically pulling semantically similar samples closer while pushing dissimilar pairs apart in the latent space, which effectively suppresses pseudochanges caused by illumination variations or seasonal shifts [45], [46], [47].

Recent works have explored various forms of contrastive learning to enhance CD performance across different supervision levels. In supervised settings, semantic contrastive objectives have been shown to improve intraclass compactness and interclass separability [48]. In unsupervised and self-supervised scenarios, contrastive learning has been used to extract robust change-sensitive features without reliance on dense annotations [42], [49], [50], [51]. Multimodal extensions have also been proposed to leverage complementary cues across spectral, temporal, or structural domains [52], further improving model robustness in challenging environments.

In contrast to existing methods that mainly apply contrastive learning during pretraining or global feature modeling, our approach introduces pixel-level contrastive learning in the later stages of supervised training, focusing on difficult examples such as small-scale changes, class boundaries, and pseudochanges caused by lighting or seasonal variations. By dynamically selecting high-uncertainty pixels and constructing contrastive pairs, our method progressively refines feature representations, enhancing the model's ability to distinguish subtle changes in complex real-world scenarios. In addition, we also introduce a change-prompt decoder (CPD), which improves detection accuracy through sparse and dense change prompts.

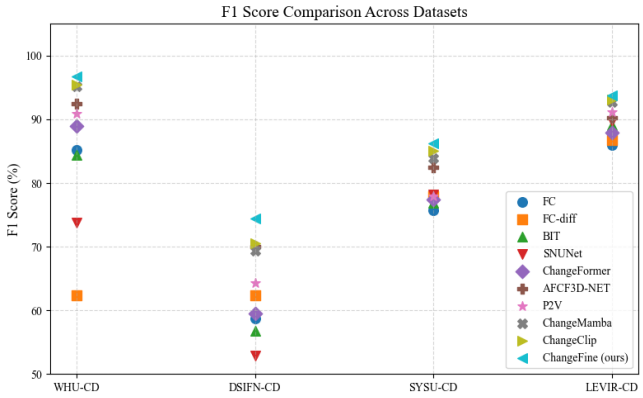


Fig. 1. *F*1 score comparison of different CD methods across five benchmark datasets (LEVIR-CD, WHU-CD, DSIFN-CD, and SYSU-CD). The proposed method, ChangeFine (ours), consistently achieves top-tier performance across all datasets, demonstrating effective generalization and robustness.

III. Proposed Method

Our proposed approach strategically exploits multimodal representations derived from foundational models (SAM and CLIP), systematically aggregating hierarchical features to achieve robust performance in CD tasks. As illustrated in Fig. 2, the framework consists mainly of four core components.

First, an SCFE is developed to jointly encode comprehensive global semantics and detailed local spatial information from both images and textual prompts, producing rich bimodal feature representations. Second, a CPFM is designed to facilitate effective interaction and alignment between bi-temporal features, significantly enhancing the discriminative capability. Finally, an HSCM is introduced to explicitly address challenging samples through contrastive learning, utilizing a dynamic memory bank to further refine change feature representations. Based on these modules, a CPD is used to fine tune the performance of the model, progressively improving the accuracy and granularity of CD by utilizing both sparse and dense change prompts.

A. SAM-CLIP Feature Encoder

To effectively integrate both global semantic understanding and fine-grained spatial information, we propose the SCFE, see Fig. 2. The motivation behind designing this module is to leverage the complementary strengths of CLIP and SAM: CLIP provides rich vision–language alignment with global semantic context, while SAM excels in extracting precise spatial structures. By integrating these two models, SCFE aims

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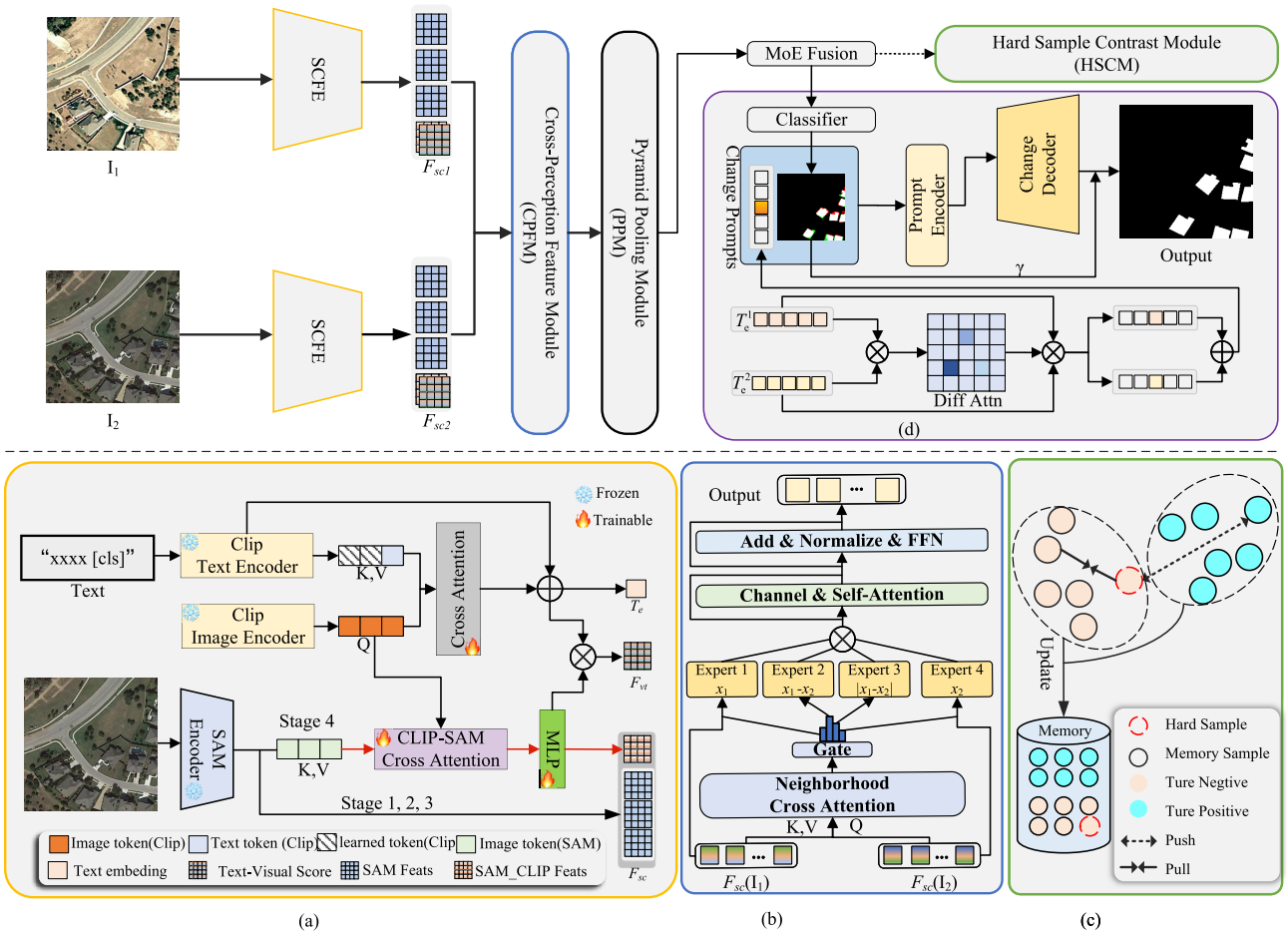


Fig. 2. Overview of the proposed *ChangeFine* framework. The (a) *SCFE* extracts joint spatial–semantic features from bi-temporal images using SAM and CLIP. The features are enhanced by the (b) *CPFM* and aggregated via a *pyramid pooling module (PPM)*. Subsequently, an optional (c) *HSCM* and (d) *CPD* are introduced to progressively refine the CD results from coarse to fine.

to generate enhanced multimodal representations that are both semantically meaningful and spatially precise, which is crucial for robust CD.

Given an input RGB image $I \in \mathbb{R}^{H\times W\times 3}$, we define a set of learnable text prompts to guide the semantic feature extraction. Inspired by context optimization (CoOp) [53], which replaces fixed manual templates with continuous learnable context vectors, the prompt $\boldsymbol{t}_m$ for the mth semantic category is formulated as

$$\boldsymbol{t}_m = [\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_L, \boldsymbol{c}_m] \tag{1}$$

where $\{\boldsymbol{v}_k\}_{k=1}^{L}$ denotes a sequence of L learnable context vectors shared across all classes to capture domain-specific spectral characteristics and $\boldsymbol{c}_m$ is the fixed word embedding of the mth class name. Based on these tokens, we construct a task-specific prompt bank $\mathcal{T} = \{\boldsymbol{t}_m\}_{m=1}^{M}$, where M is the total number of predefined categories.

The definition of these semantic categories is grounded in established hierarchical land-use classification systems and recent CD literature. The vocabulary size M is explicitly tailored to the dataset characteristics. For building-focused datasets (e.g., LEVIR-CD and WHU-CD), we employ a compact category set consisting of *building* and *background*. For complex scenes (e.g., SYSU-CD and DSIFN-CD), the vocabulary is expanded to include diverse land-cover types, such as *farmland*, *grass*, *forest*, *bare land*, *road*, and *water*. It should be emphasized that the use of multicategory prompts does not change the task formulation into multiclass CD. Instead, these prompts serve as semantic priors to enhance the category awareness of visual features. During inference, the extracted visual features are queried against all M prompts in parallel to obtain prompt-enhanced semantic representations. These representations are then passed to the subsequent feature interaction module and change decoder, which finally produces a binary changed map.

To extract high-level semantics, we encode the input image I and textual prompts T using a CLIP (ViT-B/16) [39] backbone

$$F_{cv} = f_{Img}(I), \quad F_{ct} = f_{Txt}(T) \tag{2}$$

where $F_{cv} \in \mathbb{R}^{N\times D}$ consists N local visual tokens and $F_{ct} \in \mathbb{R}^{M\times D}$ denotes the textual features for M predefined semantic categories. Here, D is the shared CLIP embedding dimension. Concurrently, we extract fine-grained spatial features using an SAM encoder: $F_{sv} = f_{SAM}(I) \in \mathbb{R}^{N\times D_s}$, where D_s is the intrinsic SAM feature dimension.

To effectively inject the global semantics of CLIP into the fine-grained spatial cues of SAM, we employ two cross-attention modules. The first module enriches the textual

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prompts with visual context by using F_{ct} as queries and the comprehensive visual features F_{cv} as keys and values

$$T_e = F_{ct} + \gamma \cdot \text{Softmax}\left(\frac{F_{ct}F_{cv}^T}{\sqrt{D/N_H}}\right)F_{cv} \tag{3}$$

yielding the image-adaptive textual representation $T_e \in \mathbb{R}^{M\times D}$, where N_H is the number of attention heads and γ is a learnable scaling parameter.

The second module explicitly injects global semantics into the spatial features. We first project the SAM features into the CLIP space via a learnable linear layer $\phi(\cdot)$, obtaining $\tilde{F}_{sv} = \phi(F_{sv}) \in \mathbb{R}^{N\times D}$. Using $\tilde{F}_{sv}$ as queries to retain spatial structures, and F_{cv} as keys and values, the refined representation $F_{scv} \in \mathbb{R}^{N\times D}$ is computed as

$$F_{scv} = \tilde{F}sv + \text{MLP}\left(\text{Softmax}\left(\frac{\tilde{F}svF_{cv}^T}{\sqrt{D/N_H}}\right)F_{cv}\right). \tag{4}$$

Subsequently, the dense visual–text similarity map is obtained via $F_{vt} = F_{scv}, T_e^T \in \mathbb{R}^{N\times M}$.

Following the ViT-adapter [54] architecture, we construct a hierarchical representation using SAM layers $l \in \{3, 5, 7, 11\}$. Features from layers 3, 5, and 7 are retained as intermediate representations. To integrate the multimodal cues seamlessly, the layer-11 feature is replaced by the projected concatenation of F_{scv} and F_{vt}

$$F_{sc} = \left\{F_{SAM}^{l=3},\ F_{SAM}^{l=5},\ F_{SAM}^{l=7},\ \psi\left(\text{Concat}\left(F_{scv}, F_{vt}\right)\right)\right\} \tag{5}$$

where $\psi(\cdot)$ is a linear projection aligning the concatenated dimension $(D + M)$ back to D_s. The resulting F_{sc} serves as the unified hierarchical representation for downstream processing.

Through the proposed SCFE, the bi-temporal images I_1 and I_2 are encoded separately as

$$F_{sc}^1 = \text{SCFE}\left(I_1\right), \qquad F_{sc}^2 = \text{SCFE}\left(I_2\right) \tag{6}$$

which provides unified bi-temporal, multimodal, and hierarchical representations for the subsequent CD network.

B. Cross-Perception Feature Module

To effectively interact and fuse the hierarchical representations produced by SCFE, we propose the CPFM, inspired by [55], to model explicit interactions between bi-temporal features. The overall workflow of CPFM is summarized in Algorithm 1. In particular, CPFM first performs local cross-temporal alignment between bi-temporal features, then constructs complementary temporal and differential cues, adaptively fuses them through a gated MoE, and finally aggregates the interacted features through a PPM to produce a robust fused representation.

The CPFM serves as a dedicated feature interaction and fusion unit that consolidates multitemporal change representations generated by the SCFE module. As illustrated in Fig. 2(b), the first attention block employs a neighborhood cross-attention mechanism (NCA), which conducts a local neighborhood search between the two temporal feature maps so that each position of the query (Q) perceives and aligns its surrounding context, thus mitigating errors that arise from residual misregistration.

Algorithm 1 CPFM

Input: Bi-temporal hierarchical features $\left\{F_{sc}^{(l,1)}, F_{sc}^{(l,2)}\right\}_{l=1}^{4}$ from SCFE.
Output: Coarse binary change prediction CD_{coarse}.
For each level $l = 1, \ldots, 4$, CPFM first aligns bi-temporal features by neighborhood cross-attention:

$$x_1^{(l)}, x_2^{(l)} \leftarrow \text{NCA}\left(F_{sc}^{(l,1)}, F_{sc}^{(l,2)}\right).$$

Then, four complementary cues are constructed, where $z_m^{(l)}$ denotes the mth cue:

$$Z^{(l)} = \left[x_1^{(l)} \parallel x_1^{(l)} - x_2^{(l)} \parallel |x_1^{(l)} - x_2^{(l)}| \parallel x_2^{(l)}\right].$$

A gated MoE adaptively fuses these cues:

$$\mathbf{g}^{(l)} = \text{Softmax}(\text{Gate}(Z^{(l)})), \quad \tilde{x}^{(l)} = \sum_{m=1}^{4} g_m^{(l)}\text{Expert}_m(z_m^{(l)}).$$

The fused feature is further refined by channel–spatial attention:

$$F^{(l)} = \text{FFN}(\text{SA}(\text{CSA}(\tilde{x}^{(l)}))) + \text{CSA}(\tilde{x}^{(l)}).$$

Finally, hierarchical features are aggregated and classified:

$$\{F_{ppm}^{(l)}\}_{l=1}^{4} = \text{PPM}(\{F^{(l)}\}_{l=1}^{4}),$$

$$\boldsymbol{\beta} = [\beta_1, \beta_2, \beta_3, \beta_4] = \text{Softmax}(\text{Gate}_s([F_{ppm}^{(1)} \parallel \cdots \parallel F_{ppm}^{(4)}])),$$

$$F_{moe} = \sum_{l=1}^{4} \beta_l F_{ppm}^{(l)}, \quad CD_{\text{coarse}} = \text{Classifier}(F_{moe}).$$

For each spatial index $i \in \{1, \ldots, N\}$, we precompute a fixed-radius neighborhood $\mathcal{N}(i) \subseteq \{1, \ldots, N\}$

$$Q_i = W_Q\left(F_{sc}^i\right), \quad K_i = W_K\left(F_{sc}^i\right), \quad V_i = W_V\left(F_{sc}^i\right) \tag{7}$$

$$\alpha_{ij} = \begin{cases} \dfrac{\exp\left(Q_i^\top K_j/\sqrt{d}\right)}{\sum_{k\in\mathcal{N}(i)}\exp\left(Q_i^\top K_k/\sqrt{d}\right)}, & j \in \mathcal{N}(i) \\ 0, & j \notin \mathcal{N}(i) \end{cases} \tag{8}$$

$$\text{NCA}\left(F_{sc}^1, F_{sc}^2\right)_i = \sum_{j\in\mathcal{N}(i)} \alpha_{ij} V_j. \tag{9}$$

Here, $W_q, W_k, W_v \in \mathbb{R}^{d\times d}$ are learnable projections and d is the embedding dimension. The neighborhood mask ensures that each Q Q_i interacts only with keys and values inside its local window $\mathcal{N}(i)$, thus aligning fine-grained contexts and reducing residual misregistration.

After the NCA stage has locally aligned the bi-temporal features, the CPFM inserts a layer *gated mixture of experts* (MoE [56]) to adaptively fuse differential cues. Concretely, four parallel expert branches are instantiated at every spatial location to process the first temporal features x_1, the raw difference $x_1 - x_2$, the absolute difference $|x_1 - x_2|$, and

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the second temporal features x_2. Retaining both x_1 and x_2 explicitly allows the model to preserve essential temporal semantics that may not be captured by difference operations alone. This helps the gating mechanism distinguish meaningful changes from noise or background fluctuations. A lightweight gating network then predicts pixelwise, normalized weights $g_m \in [0,1]$ with $\sum_{m=0}^{3} g_m = 1$ based on the local semantic context, and outputs the fused representation

$$\tilde{x} = \sum_{m=0}^{3} g_m \, \text{Expert}_m(\cdot). \tag{10}$$

This dynamic MoE scheme accentuates genuine change regions, suppresses background or spurious differences, and provides a more discriminative representation for the subsequent channel self-attention and feed-forward network.

After MoE fusion, we obtain the change-aware tensor $\tilde{x} \in \mathbb{R}^{H\times W\times d}$, which is refined by a *channel–spatial transformer block* that first applies channel self-attention to reweight inter-channel responses and then feeds the result into a multihead spatial self-attention followed by a residual FFN, yielding

$$F^1 = \text{FFN}(\text{SA}(\text{CSA}(\tilde{x}))) + \text{CSA}(\tilde{x}) \tag{11}$$

where CSA(·) and SA(·) denote channel and spatial self-attention, respectively.

The same operation is applied to features at other stages. To further aggregate these hierarchical representations, we employ a PPM to generate the final multilevel fused representation F_{ppm}

$$F^l_{ppm} = \text{PPM}\left(\left\{F^{(l)}\right\}_{l=1}^{4}\right). \tag{12}$$

To adaptively fuse multiscale feature representations F^l_{ppm}, we propose another scale-aware MoE fusion module. Given the four scale-level features $\{F^{(1)}_{ppm}, F^{(2)}_{ppm}, F^{(3)}_{ppm}, F^{(4)}_{ppm}\}$, each is first normalized and reshaped into patchwise token sequences. These representations are concatenated and passed through a lightweight gating network to compute spatially adaptive weights over the four expert branches. The fused feature F_{moe} is then computed as a weighted summation of the expert features

$$F_{moe} = \sum_{i=1}^{4} \beta_i \cdot F^{(i)}_{ppm} \tag{13}$$

$$\boldsymbol{\beta} = \text{Softmax}\left(\text{Gate}\left(\left[F^{(1)}_{ppm} \| F^{(2)}_{ppm} \| F^{(3)}_{ppm} \| F^{(4)}_{ppm}\right]\right)\right). \tag{14}$$

Here, $\boldsymbol{\beta} \in \mathbb{R}^{B\times N\times 4}$ denotes the learned importance weights for each expert at each spatial position, and $\|$ denotes the concatenation of features along the dimension of the channel. The fused representation is finally projected to the target embedding dimension via a linear transformation. This design enables the network to dynamically emphasize the most informative scales based on local context, enhancing its robustness to scale variations and semantic diversity in complex scenes. Subsequently, we obtain the coarse prediction CD_{coarse} as follows:

$$CD_{coarse} = \text{Classifier}(F_{moe}) \tag{15}$$

where Classifier(·) is implemented as a 1×1 convolutional layer, which transforms the fused feature F_{moe} into pixelwise change logits and outputs the coarse prediction map.

C. Change-Prompt Decoder

To achieve coarse-to-fine CD, we propose a change-prompt-guided detector that synergistically combines implicit sparse change and explicit dense change cues.

As shown in Fig. 2(d), the initial extraction of the implicit sparse change signal is achieved by evaluating the differences between the textual embeddings T^1_e and T^2_e produced by the SCFE module (refer to Section III-A). The sparse change prompt of text is determined as follows:

$$\text{DiffAttn} = 1 - \text{softmax}\left(T^1_e \cdot \left(T^2_e\right)^{\top}\right)$$
$$T_e = \left(T^1_e + T^2_e\right) \otimes \text{DiffAttn} \tag{16}$$

where $\otimes$ signifies elementwise multiplication. This process enhances tokens associated with semantically divergent regions while attenuating redundant or unchanged areas, thereby producing a cohesive semantic representation T_e that functions as a sparse prompt.

Subsequently, the dense change prompt is implemented through CD_{coarse} and the image feature F_{moe}, as elaborated in Section III-B.

To enhance the adaptability of the pretrained (SAM) for downstream fine-grained change prediction. Fig. 3 shows a PEFT strategy based on low-rank adaptation (LoRA) [26] within the two-way Transformer module. In particular, the original Q-key–value (K/V) projection weights (W_Q, W_K, W_V) of the transformer are *frozen*, while LoRA injects trainable low-rank matrices (A, B) to form

$$W' = W + \alpha BA \tag{17}$$

where $W \in \mathbb{R}^{d\times d}$ is the frozen pretrained weight, $A \in \mathbb{R}^{r\times d}$ and $B \in \mathbb{R}^{d\times r}$ are the trainable low-rank factors, $r \ll d$ denotes the rank of the adaptation, and α is a scaling factor that controls the update magnitude. During training, only the parameters A and B (along with a small number of normalization layers) are optimized, while all other parameters remain frozen. The total number of parameters in this part is 3.34M, of which only 0.05M are trainable.

In conclusion, the explicit and implicit change prompts are integrated within an adapted SAM decoder to produce the ultimate change prediction

$$CD_{fine} = \lambda \cdot \text{SAM}_{Decoder}(F_{moe}, T_e, CD_{coarse}) + (1-\lambda) \cdot \text{CD}_{coarse} \tag{18}$$

where λ is a learnable parameter. This progressive refinement strategy effectively integrates both implicit and explicit guidance to achieve fine-grained and accurate change localization. By jointly utilizing the tripartite architecture, our decoder progressively refines the detection outcomes, capturing subtle and localized changes while maintaining high robustness and precision.

D. Hard Sample Contrast Learning Module

In CD tasks, hard samples, particularly at the pixel level, are crucial for accurately distinguishing subtle and ambiguous changes from background noise. Such hard samples typically correspond to regions characterized by low contrast, partial

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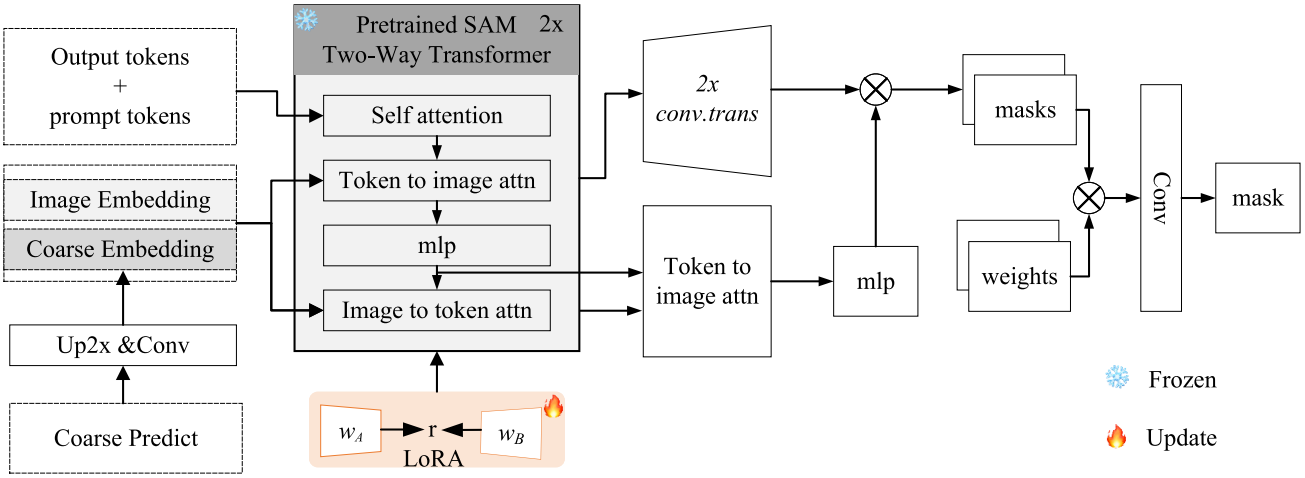


Fig. 3. Architecture of the CPD.

occlusions, or mixed spectral characteristics, which makes it inherently challenging to classify accurately. To effectively handle these challenging samples, we introduce a specific strategy combining hard sample selection with contrastive learning, enhancing model robustness and generalization.

The overall process of HSCM is illustrated in Fig. 4. False positive (FP)/ false negative (FN) pixels are selected as hard anchors and contrasted with the stored changed/unchanged feature representations in the memory bank, after which the refined features are used to update the memory bank. In particular, as shown in Fig. 2(c), during the initial phase of training, we store pixel-level feature representations F_{moe} in a memory bank, maintaining separate feature sets for changed and unchanged pixels. At the beginning of the training iterations, we strategically select a set of challenging pixels, consisting of FPs and FN pixels, as hard sample anchors. Once hard sample anchors are selected, we perform pixel-level contrastive learning. Each hard sample anchor is contrasted against easy positive and negative pixel features retrieved from the memory bank. Through the explicit contrast between difficult samples and representative easy samples, the model learns to embed these challenging pixels into appropriate semantic clusters. Formally, the contrastive learning process is defined by the following contrastive loss function:

$$\mathcal{L}_{con} = -\frac{1}{N_h}\sum_{i=1}^{N_h}\log\frac{\sum_{j\in P_i^e} e^{s(h_i,p_j)/\tau}}{\sum_{j\in P_i^e} e^{s(h_i,p_j)/\tau} + \sum_{k\in N_i^e} e^{s(h_i,n_k)/\tau} + \epsilon} \tag{19}$$

where N_h denotes the number of selected hard anchors and h_i denotes the ith hard anchor. P_i^e is the set of easy positive samples sharing the same class label as h_i, while N_i^e denotes the set of easy negative samples from the opposite class. p_j and n_k represent the corresponding positive and negative samples, respectively, and $s(\cdot,\cdot)$ denotes the cosine similarity between two features. τ is the temperature parameter, and ϵ is a small constant for numerical stability. In this way, the loss explicitly focuses on hard anchors while using relatively reliable easy samples as references, which helps the model learn more discriminative representations for ambiguous regions.

Through this hard-sample-aware contrastive learning framework, the model explicitly addresses ambiguous and difficult-to-classify regions, significantly enhancing feature discriminability and improving accuracy in pixel-level CD tasks.

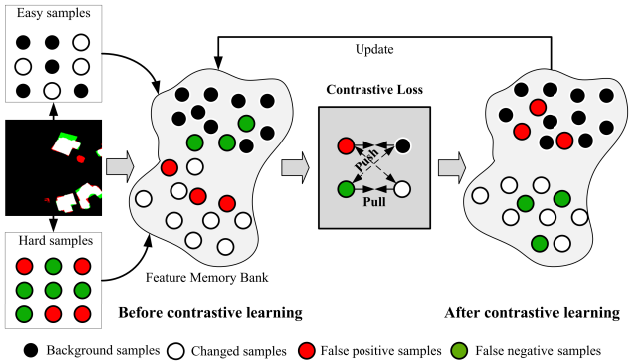


Fig. 4. Illustration of the proposed HSCM.

E. Loss Functions

To effectively optimize the proposed method, we adopt a hybrid loss function that integrates multiple supervision signals tailored to different stages and components of the network. The general objective is to improve the model's ability to capture semantic consistency, distinguish hard samples, and achieve fine-grained change localization. In particular, we employ a segmentation loss to guide pixel-level classification, a contrastive loss to enhance feature discriminability for hard pixels, and an auxiliary loss to facilitate multilevel feature learning during training. The total loss is defined as

$$\mathcal{L}_{total} = \mathcal{L}_F + \mathcal{L}_c + \lambda_{con}\mathcal{L}_{con}. \tag{20}$$

Here, $\mathcal{L}_F$ and $\mathcal{L}_c$ denote the fine- and coarse-grained segmentation losses, respectively, both composed of a combination of focal loss [57] and Dice loss [58] applied at different stages of the network. $\mathcal{L}_{con}$ is the contrastive loss designed to emphasize hard sample discrimination, and λ_{con} is an empirically set balancing coefficient of 0.1 in our experiments.

IV. Experiment Details

In this chapter, to thoroughly evaluate the performance and generalization ability of the ChangeFine network, we

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conducted experiments on four challenging publicly available CD datasets. We provide a detailed description of the experimental setup, evaluation metrics, and the processing details for each dataset. Subsequently, we compared our method with classical CD techniques and state-of-the-art methods based on different architectures. Both quantitative results and visual outcomes are thoroughly analyzed.

A. Implementation Details

All experiments were conducted using a single NVIDIA GeForce RTX 3090 GPU within the PyTorch framework. To ensure fair comparisons, all methodologies were trained over 200 epochs with a batch size of 24. The Adam [59] optimizer was utilized, starting with an initial learning rate of $2e^{-5}$, which was gradually decreased to $1e^{-8}$ according to a cosine annealing schedule. The weight decay parameter was consistently set at 0.01. To enhance the stability of training during the initial stages and to reduce overfitting, the HSCM was activated after the 75% epoch. Data augmentation included random horizontal flipping, rotation, resizing, and saturation adjustment.

B. Datasets

In this study, to fully verify the performance and generalization of our model on different datasets. We used four publicly available datasets for the experiment, which are described as follows.

The WHU-CD [60] dataset is a publicly available building CD dataset, which consists of two high-resolution aerial images with a size of 32 507 × 15 354 and is widely used in building CD research. In order to ensure the fairness of the experiment, we adopted a segmentation scheme consistent with previous studies, cutting the image into 256 × 256 pixels without overlap, and randomly dividing it into three parts: 6096/762/762, which were respectively used for training/verification/testing.

The SYSU-CD [33] dataset is a public CD dataset taken in Hong Kong. Besides building changes, it also includes urban expansion, road changes, coastal changes, and vegetation changes. It is an important remote-sensing image resource for studying coastal urban changes. The dataset consists of 20 000 pairs of high-resolution aerial images with sizes of 256 × 256 pixels, 12 000 for training, 4000 for validation, and 4000 for testing.

The DSIFN-CD [34] dataset consists of high-resolution remote-sensing images of six Chinese cities: Beijing, Chengdu, Shenzhen, Chongqing, Wuhan, and Xi'an. The remote-sensing images of Beijing, Chengdu, Shenzhen, Chongqing, and Wuhan are used for training and verification, and the remote-sensing images of Xi'an are used for testing. We took the cropping scheme consistent with the previous research, cropping the 512 × 512 pixel size image into 256 × 256 small pieces, and divided it into three parts: 14400/1360/192, respectively, training/validation/testing.

The LEVIR-CD [11] dataset is a publicly available benchmark for building CD in high-resolution remote-sensing imagery. It contains 637 bi-temporal image pairs collected from various urban areas, with each pair covering a spatial extent of 1024 × 1024 pixels. The images were sourced from Google Earth and captured at different times spanning a period of 5 to 14 years. Each pair is annotated with a pixel-level binary change mask indicating building changes. Following previous works, we cropped each image into nonoverlapping 256 × 256 patches and divided the dataset into three subsets: 7120 image pairs for training, 1024 for validation, and 2048 for testing.

C. Evaluation Metrics

We use six metrics, Precision, Recall, mIoU, OA, IoU, and $F1$ score, to comprehensively and quantitatively evaluate the performance of remote-sensing image CD models. Among them, IoU and $F1$ score are the main indicators for assessing the CD results. Higher evaluation indicators mean better prediction results. The definitions of the evaluation metrics are as follows:

$$\begin{aligned} Precsion &= \frac{TP}{TP+FP} \\ Recall &= \frac{TP}{TP+FN} \\ IoU &= \frac{TP}{TP+FN+FP} \\ F1 &= \frac{2\,(Precsion \times Recall)}{Precsion + Recall} \\ OA &= \frac{TP+TN}{TP+TN+FP+FN} \\ mIoU &= \frac{1}{N}\sum_{n=1}^{N}\frac{TP_n}{TP_n+FP_n+FN_n} \end{aligned} \tag{21}$$

where TP represents the number of true positives (TPs), TN represents the number of true negatives, FP represents the number of false positives, and FN represents the number of false negatives.

D. Comparison With State-of-the-Art Methods

To fully validate ChangeFine's performance and advances, we compare eight CD methods, including: FC-EF [9], BIT [32], SNUNet [61], ChangeFormer [13], AFCF3D-Net [30], P2V [62], SAM-CD [22], ChangeCLIP [20], and ChangeMamba [41], and give a brief introduction to each method.

The FC-EF [9] is a fully convolutional U-Net-based architecture for CD. It concatenates two input images as channels and processes them through a shallower U-Net (4 pooling/upsampling layers).

The BIT [32] is a hybrid CNN-Transformer model for CD, using semantic tokenizers to extract concepts and transformer encoders–decoders to model global context and refine features.

The SNUNet [61] is a Siamese encoder–decoder network for CD, featuring dense skip connections to preserve spatial details and an ensemble channel attention module (ECAM) to fuse multilevel features effectively.

The ChangeFormer [13] is a transformer-based CD network featuring a hierarchical encoder, multiscale difference modules, and an MLP decoder to fuse features and predict change maps efficiently.

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TABLE I
CD RESULTS ON WHU-CD AND LEVIR-CD DATASETS. (TOP-3 MARKED BY RED/BLUE/GREEN)

Method	WHU-CD						LEVIR-CD						Params
	Prec.	Rec.	mIoU	OA	IoU	F1	Prec.	Rec.	mIoU	OA	IoU	F1	
FC-EF	90.99	84.51	88.43	98.91	77.99	87.63	89.57	87.29	89.01	98.83	79.24	88.42	1.35
BIT	81.34	90.95	86.91	98.63	75.25	85.88	91.77	89.40	90.89	99.05	82.77	90.57	3.04
SNUNET	62.93	92.17	78.38	97.16	59.73	74.79	91.82	89.74	91.06	99.07	83.10	90.77	12.03
ChangeFormer	92.89	86.34	90.01	99.07	80.99	89.50	90.92	86.03	89.00	98.85	79.22	88.41	41.03
AFCF3D-NET	96.52	88.79	92.67	99.34	86.04	92.49	92.58	87.93	90.56	99.03	82.13	90.19	17.54
P2V	92.64	88.29	90.80	99.14	82.50	90.41	92.23	89.09	90.95	99.06	82.88	90.64	5.42
SAM-CD	96.06	89.06	92.62	99.33	85.93	92.43	90.73	86.37	89.09	98.86	79.37	88.50	2.49
ChangeMamba	96.18	92.23	94.30	99.58	89.02	94.19	92.86	90.80	91.95	99.17	84.78	91.76	84.70
ChangeClip	95.58	92.42	94.10	99.50	88.85	93.96	92.29	90.90	91.80	99.15	84.49	91.59	120.83
ChangeFine	96.00	93.50	94.75	99.52	90.00	94.73	92.61	91.17	92.06	99.18	84.99	91.89	69.57

TABLE II
CD RESULTS ON SYSU-CD AND DSIFN-CD DATASETS. (TOP-3 MARKED BY RED/BLUE/GREEN)

Method	SYSU-CD						DSIFN-CD					
	Prec.	Rec.	mIoU	OA	IoU	F1	Prec.	Rec.	mIoU	OA	IoU	F1
FC-EF	79.57	77.02	75.99	89.92	64.30	78.27	57.05	66.22	64.09	85.79	44.19	61.29
BIT	80.41	76.44	76.15	90.05	64.44	78.38	52.35	65.62	61.54	84.01	41.08	58.24
SNUNET	81.39	77.10	76.93	90.44	65.54	79.18	41.37	77.07	55.50	77.54	36.84	53.84
ChangeFormer	82.22	74.12	75.95	90.12	63.88	77.96	66.19	54.87	64.58	87.57	42.86	60.02
AFCF3D-NET	82.73	82.21	79.97	91.76	70.17	82.47	65.14	75.63	70.60	88.98	53.84	69.96
P2V	77.36	77.56	75.09	89.36	63.21	77.46	61.11	66.66	66.14	87.13	46.81	63.77
SAM-CD	83.14	74.43	76.52	90.41	64.67	78.55	66.76	63.56	67.65	88.43	48.28	65.12
ChangeMamba	87.70	78.67	80.74	92.37	70.86	82.94	75.69	62.38	70.50	90.20	51.97	68.39
ChangeClip	84.98	82.12	81.12	92.36	71.71	83.53	65.10	73.50	69.96	88.80	52.72	69.05
ChangeFine	82.33	86.09	81.80	92.69	72.67	84.17	67.22	78.41	72.49	89.83	56.72	72.38

The AFCF3D-Net [30] is a 3-D CNN-based CD network featuring adjacent-level feature cross-fusion (AFCF) modules and a full-scale decoder to effectively integrate spatiotemporal features from bi-temporal images.

The P2V [62] is a dual-branch CD network that processes bi-temporal images both as spatial pairs and temporal pseudovideos, using separate encoders for spatial and temporal feature extraction with lateral connections.

The ChangeCLIP [20] is a vision–language CD framework that integrates CLIP's multimodal capabilities with a differential feature compensation module and vision–language decoder to enhance CD in remote-sensing images.

The ChangeMamba [41] is a Mamba architecture-based framework for RSCD, featuring visual SSMs and cross-scan mechanisms to efficiently model spatiotemporal relationships in bi-temporal images.

The SAM-CD [22] is a framework adapting the SAM for CD, utilizing a fixed FastSAM encoder to extract visual representations and a learnable convolutional adaptor to aggregate task-specific change information from VHR images.

We chose methods with different architectures that are representative in the CD field that achieve greater results as comparison methods. For a fair comparison, all baseline methods were implemented based on their publicly released source codes and retrained under the same experimental protocol as our method. In particular, all methods used the same dataset splits, input resolution, data augmentation strategy, evaluation metrics, and random seed, ensuring strictly consistent training and testing conditions across different models.

E. Quantify Analysis and Visualize Results

As illustrated in Tables I and II, our proposed ChangeFine method consistently outperforms comparative state-of-the-art methods, highlighting its robust accuracy and robustness across diverse remote-sensing datasets. Below, we provide detailed quantitative and qualitative analyses organized by dataset.

Experimental Results on the WHU-CD Dataset: The left side of Table I presents the quantitative results on the WHU-CD dataset. Our method, ChangeFine, achieves the highest performance across the key evaluation metrics, including IoU, $F1$ score, mIoU, and Recall. In particular, ChangeFine achieves an IoU of 90.00%, an $F1$ score of 94.73%, and an mIoU of 94.75%. In comparison, ChangeMamba, the closest competitor, achieves an IoU of 89.02%, an $F1$ score of 94.19%, and an mIoU of 94.30%. ChangeFine outperforms

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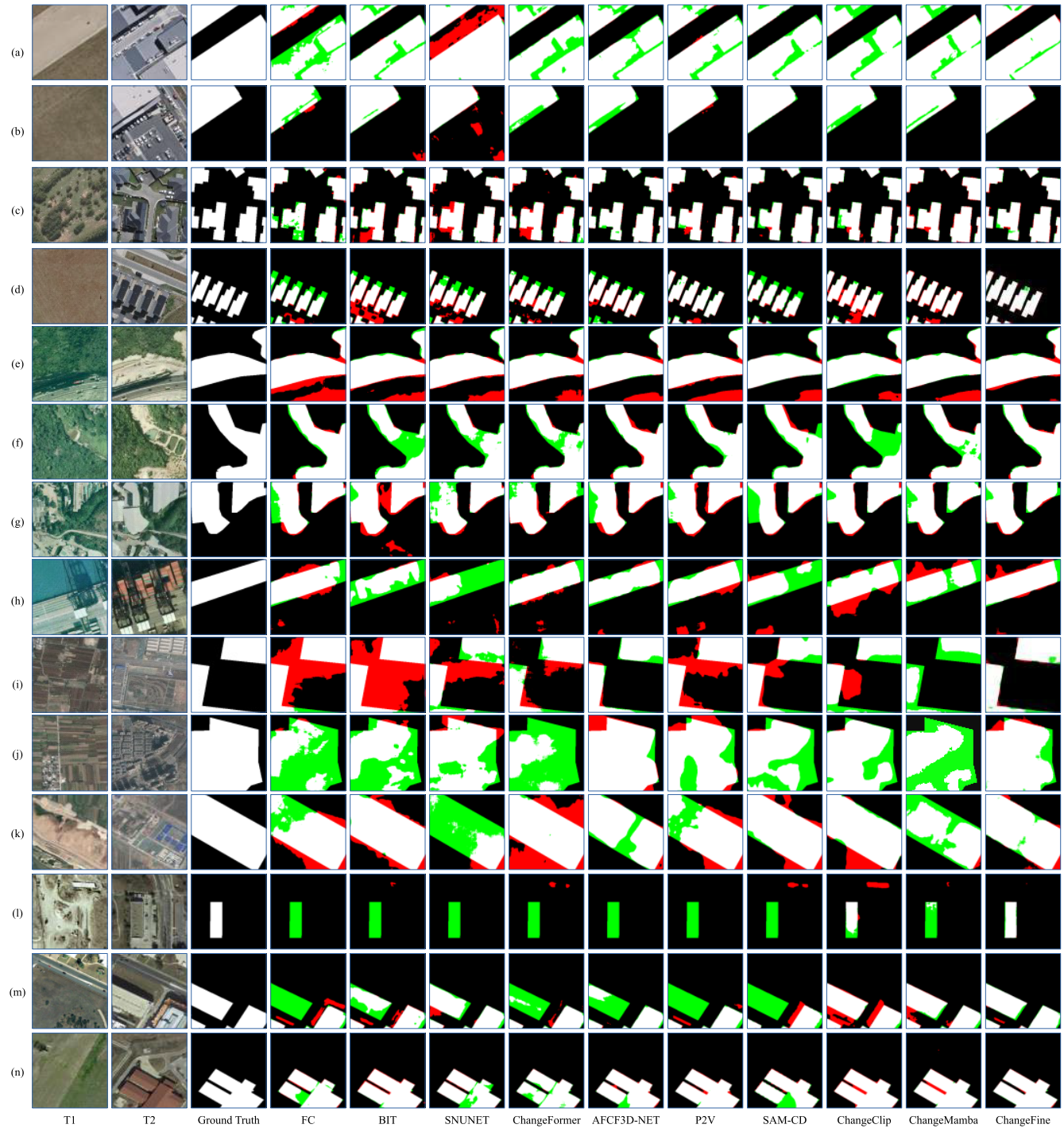


Fig. 5. Qualitative comparison results on four datasets: (a)–(d) WHU-CD, (e)–(h) SYSU-CD, (i)–(k) DSIFN-CD, and (l)–(n) LEVIR-CD. In the visualization, TP are shown in white, FP in red, and false negatives (FN) in green.

ChangeMamba by 0.98% in IoU, 0.54% in $F1$ score, and 0.45% in mIoU. Furthermore, ChangeFine significantly outperforms other methods. It improves IoU by 14.75% over BIT, 4.07% over SAM-CD, and 1.15% over ChangeClip. In terms of $F1$ score, ChangeFine surpasses BIT by 8.85%, SAM-CD by 2.30%, and ChangeClip by 0.77%. These results highlight ChangeFine's exceptional performance in CD, with outstanding results across key comprehensive metrics, particularly in terms of $F1$ score and IoU.

Fig. 5(a)–(d) shows the visualization results for the WHU-CD dataset. Each row displays the CD results for different scenes, while the columns show the outputs from different methods. Fig. 5(a) and (b) illustrate large-scale changes in buildings. ChangeFine effectively captures the boundaries of the changed areas with minimal misclassification. In contrast, ChangeMamba, ChangeClip, and SAM-CD exhibit shortcomings in boundary identification, with noticeable FPs and false negatives, particularly in areas of large-

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scale changes. Fig. 5(c) and (d) demonstrate the detection results for dense buildings, where ChangeFine precisely delineates the changed areas. Compared to SAM-CD and ChangeClip, ChangeFine significantly reduces FPs and false negatives (FN), showing better object completeness. This analysis clearly shows that ChangeFine offers more reliable and robust performance across all scenes, particularly in complex and subtle CD, effectively minimizing detection omissions while maintaining state-of-the-art $F1$ score and IoU performance.

Experimental Results on the LEVIR-CD Dataset: The right side of Table I displays the evaluation results on the challenging LEVIR-CD test dataset. It is evident that ChangeFine surpasses other methodologies across several critical metrics, including Recall, mIoU, OA, $F1$, and IoU. In particular, ChangeFine achieves the highest $F1$ score of 91.89%, an IoU of 84.99%, and an mIoU of 92.06%. Compared to ChangeMamba, ChangeFine shows improvements of 0.13% in $F1$ score, 0.21% in IoU, and 0.11% in mIoU. Notably, ChangeFine significantly outperforms SAM-CD, achieving a 5.62% higher IoU and a 3.39% higher $F1$ score. These results show that ChangeFine achieves a consistent accuracy improvement over ChangeClip, ChangeMamba, and SAM-CD on the LEVIR-CD dataset, particularly evident in complex urban scenes.

Fig. 5(l)–(n) displays the CD results for the LEVIR-CD dataset. From Fig. 5(l), we observe that when buildings and their surrounding areas have similar colors, most methods tend to miss the changed targets. However, ChangeFine performs better than SAM-CD and ChangeClip, showing fewer FP and false negatives (FN). While SAM-CD generally detects the targets, ChangeFine detects the target more completely, with results closely aligned with the ground truth. In Fig. 5(m) and (n), we find that ChangeFine excels in detecting subtle details and boundaries where changes are harder to distinguish. The detected change area closely matches the actual change region, whereas SAM-CD, ChangeMamba, and ChangeClip introduce some FPs.

Experimental Results on the SYSU-CD Dataset: The left side of Table II provides the quantitative results obtained from the SYSU-CD dataset. ChangeFine exhibits highly competitive performance across the key evaluation metrics, including IoU, $F1$ score, Recall, OA, and mIoU. In particular, ChangeFine attains an IoU of 72.67%, an $F1$ score of 84.17%, and an mIoU of 81.80%. In contrast, the nearest competitor, ChangeClip, achieves an IoU of 71.71%, an $F1$ score of 83.53%, and an mIoU of 81.12%. ChangeFine surpasses ChangeClip by 0.96% in IoU, 0.64% in $F1$ score, and 0.68% in mIoU, illustrating its enhanced capability to localize change regions with significantly higher recall and overall accuracy. In addition, ChangeFine exhibits significant improvements over other methods. It improves IoU by 8.23% relative to BIT, 9.46% over P2V, and 8.00% over SAM-CD. Regarding the $F1$ score, ChangeFine surpasses BIT by 5.79%, AFCF3D-NET by 1.70%, and SAM-CD by 5.62%. These results underscore ChangeFine's exceptional performance in CD, exhibiting a reliable $F1$ score and IoU compared to recent state-of-the-art methods like SAM-CD and ChangeMamba.

Fig. 5(e)–(h) presents the visualization outcomes for the SYSU-CD dataset. Fig. 5(e) and (f) illustrates large-scale terrain changes, where the boundaries of the changed areas are well-defined, and misclassification regions are minimal with ChangeFine. In contrast, ChangeMamba, SAM-CD, and ChangeClip are deficient in boundary identification, manifesting considerable FPs and FNs. In particular, SAM-CD tends to miss detailed boundary information in these regions. Fig. 5(g) and (h) shows the detection results for dense building areas. ChangeFine proficiently delineates the changed areas, while SAM-CD and ChangeClip struggle with separating closely spaced changes. ChangeFine significantly diminishes false negatives (FN) and captures more complete change regions, demonstrating advantageous change localization and boundary integrity in these intricate regions.

This assessment indicates that ChangeFine consistently delivers more consistent and robust performance in all scenarios, particularly in detecting complex and subtle changes.

Experimental Results on the DSIFN-CD Dataset: The right side of Table II presents an assessment performed on the challenging DSIFN-CD test dataset, elucidating the specific performance attributes of diverse CD methodologies. In particular, our proposed ChangeFine approach exhibits outstanding performance, securing the highest scores across several key metrics: Recall 78.41%, mIoU 72.49%, IoU 56.72%, and $F1$ score 72.38%. This well-balanced performance profile indicates that ChangeFine effectively addresses the intricate challenges intrinsic to large-scale bi-temporal satellite imagery analysis. Compared to other methodologies, ChangeFine demonstrates better performance compared to approaches such as ChangeMamba, ChangeClip, and SAM-CD. In particular, ChangeFine achieves an IoU of 56.72%, which is an enhancement of 4.75% over ChangeMamba's 51.97% and exhibits further superiority over SAM-CD, with an IoU advancement of 8.44% and a corresponding increase in $F1$ score of 7.26%. Fig. 5(i)–(k) displays the CD results for the DSIFN-CD dataset. We observe that, even in complex changes with unclear boundaries, ChangeFine maintains the highest similarity with the ground truth. While SAM-CD captures the general change area, it lacks the boundary precision seen in ChangeFine. Compared to AFCF3D-NET and SAM-CD, ChangeFine excels in the completeness of the detected objects, showcasing the advantage of our method in preserving target integrity.

Model Complexity Analysis; As shown in the Params column of Table I, ChangeFine contains 69.57-Mb parameters. Although it is larger than early lightweight models such as BIT (3.04 Mb) and FC-EF (1.35 Mb), this moderate increase is justified by the clear performance gains across all benchmark datasets. More importantly, compared with recent state-of-the-art methods, ChangeFine is more parameter-efficient, being 17.8% smaller than ChangeMamba (84.70 Mb) and 42.4% smaller than ChangeClip (120.83 Mb), while still achieving better IoU and $F1$ score.

This efficiency mainly comes from our parameter-efficient design. During training, the heavy CLIP and SAM encoders are kept frozen, and only the lightweight interaction and decoding modules are optimized. In addition, a LoRA-based

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TABLE III
IMPACT OF SCFE ON DIFFERENT DATASETS. EACH SCORE IS REPORTED AS IOU/$F1$(%)

Encoder	Dataset			
	SYSU-CD	DSIFN-CD	WHU-CD	LEVIR-CD
CLIP Only	71.26/83.21	54.14/70.25	87.06/93.08	82.21/90.26
SAM Only	72.03/83.74	50.61/67.20	87.41/93.28	83.93/91.26
SCFE (SAM as Q)	72.14/83.82	54.36/70.44	88.71/94.10	84.01/91.32
SCFE (CLIP as Q)	**72.67/84.17**	**56.72/72.38**	**90.00/94.73**	**84.99/91.89**

TABLE IV
ABLATION RESULTS OF THE PROPOSED MODULES ON FOUR DATASETS. EACH ENTRY REPORTS IOU/$F1$ (%)

Configuration	LEVIR-CD	SYSU-CD	DSIFN-CD	WHU-CD
Baseline	81.84/90.01	67.73/80.76	53.27/69.51	86.75/92.76
Baseline + CPFM	82.98/90.53	69.78/82.20	54.77/70.77	87.68/93.44
Baseline + CPFM + CPD	83.52/91.02	71.86/83.63	55.64/71.50	88.89/94.12
Full Model	**84.99/91.89**	**72.67/84.17**	**56.72/72.38**	**90.00/94.73**

low-rank adaptation strategy is adopted in the decoder to enhance task-specific learning with minimal parameter overhead. Therefore, the performance gains of ChangeFine come from effective architectural design rather than simply increasing model size. These results demonstrate that ChangeFine achieves a favorable tradeoff between detection accuracy and model complexity.

Based on the comprehensive analysis above, ChangeFine consistently outperforms both early convolutional benchmarks and recent state-of-the-art methods across multiple datasets, including WHU-CD, SYSU-CD, DSIFN-CD, and LEVIR-CD. By excelling in key metrics such as IoU, $F1$ score, and Recall, the proposed method demonstrates robustness in diverse scenarios. Notably, ChangeFine achieves an optimal balance between model complexity and detection accuracy; while it possesses the necessary capacity to significantly surpass early lightweight models, it remains substantially more parameter-efficient than recent heavyweight architectures like ChangeMamba and ChangeClip. Its ability to effectively detect both large-scale terrain changes and fine-grained structural details highlights its strong reliability and comprehensive detection accuracy.

V. ABLATION STUDY

To thoroughly evaluate the individual contribution and effectiveness of each core component in the proposed method, we first conduct progressive ablation experiments on the overall architecture across four highly diverse benchmark datasets, namely, LEVIR-CD, SYSU-CD, DSIFN-CD, and WHU-CD. Subsequently, we select the LEVIR-CD dataset to perform a fine-grained analysis of the internal parameter configurations of the key submodules.

A. Impact of SCFE

Table III illustrates the effectiveness of the SCFE module on four benchmark change-detection datasets (SYSU-CD, DSIFN-CD, WHU-CD, and LEVIR-CD), where each entry denotes IoU/$F1$ (%). Compared with the best single-encoder baselines ("CLIP Only" and "SAM Only"), SCFE consistently improves both intersection over union and $F1$ scores across all datasets, confirming its effectiveness in enhancing cross-modal feature interactions.

In particular, on SYSU, SCFE (CLIP as Q) attains 72.67/84.17, outperforming SAM-only by 0.64/0.43. For the more challenging DSIFN dataset, SCFE (CLIP as Q) yields 56.72/72.38, improving upon CLIP-only by 2.58/2.13. On WHU-CD, the IoU increases from 87.41 to 90.00 (+2.59), demonstrating that SCFE facilitates more accurate boundary alignment. For LEVIR-CD, the proposed model still gains +1.06 IoU and +0.63 $F1$.

Using CLIP features as the Q yields the highest performance because CLIP encodes stable high-level semantics that guide the selection of relevant visual responses from the SAM encoder's K/V features. These results validate that the SCFE module injects a semantic-guided Q branch into cross-attention, allowing the model to focus on semantically consistent regions when aligning heterogeneous representations. These results validate the effectiveness of combining the complementary capabilities of SAM and CLIP for improved semantic–spatial representations.

B. Impact of Different Modules

Since SCFE (w/CLIP as Q) demonstrates the best tradeoff between semantic robustness and spatial precision in the above comparative experiments, it is adopted as the backbone feature extractor in the subsequent ablation studies. Based on this setting, we further investigate the individual contributions and overall effectiveness of the remaining core components in the proposed framework, namely, CPFM, CPD, and HSCM. Table IV reports the progressive ablation results of the proposed modules on four benchmark datasets. Starting from the baseline built upon SCFE, introducing CPFM consistently improves performance across all datasets, confirming its effectiveness in enhancing cross-modal perception and contextual interaction. Further incorporating CPD brings additional gains, showing that the coarse-to-fine decoding strategy can effectively refine change representations. With the final integration of HSCM, the full model achieves the best results on all four datasets, reaching 84.99/91.89 on LEVIR-CD, 72.67/84.17 on SYSU-CD, 56.72/72.38 on DSIFN-CD, and 90.00/94.73 on WHU-CD. Compared with the baseline, the full model improves IoU/$F1$ by 3.15/1.88, 4.94/3.41, 3.45/2.87, and 3.25/1.97, respectively. These results demonstrate that CPFM, CPD, and HSCM provide complementary benefits and jointly enhance the robustness and discriminative capability of the proposed framework.

C. Impact of CPD

To evaluate the effectiveness of prompts, a comparative analysis is performed in scenarios contrasting with and without prompt guidance. As indicated in Table V, the introduction of coarse prompts consistently enhances performance in four benchmark datasets: LEVIR-CD, WHU-CD, DSIFN, and

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TABLE V
IMPACT OF PROMPT

Dataset	Type	Precision	Recall	IOU	OA	F1
LEVIR-CD	w/o prompts	90.72	90.00	82.41	**99.19**	90.78
	w prompts	**92.61**	**91.17**	**84.99**	99.18	**91.89**
WHU-CD	w/o prompts	93.81	**93.52**	88.35	99.43	93.88
	w prompts	**96.00**	93.50	**90.00**	**99.52**	**94.73**
DSIFN-CD	w/o prompts	65.90	77.47	55.30	89.36	71.52
	w prompts	**67.22**	**78.41**	**56.72**	**89.83**	**72.38**
SYSU-CD	w/o prompts	**83.31**	84.07	71.95	92.27	83.69
	w prompts	82.33	**86.09**	**72.67**	**92.69**	**84.17**

TABLE VI
IMPACT OF HSCM

Dataset	Type	Precision	Recall	IOU	OA	F1
LEVIR-CD	w/o contrast	91.47	90.58	83.52	99.09	91.02
	w contrast	**92.61**	**91.17**	**84.99**	**99.18**	**91.89**
WHU-CD	w/o contrast	96.72	91.66	88.89	99.48	94.12
	w contrast	**96.00**	**93.50**	**90.00**	**99.52**	**94.73**
DSIFN-CD	w/o contrast	70.31	72.73	55.64	90.15	71.50
	w contrast	**67.22**	**78.41**	**56.72**	**89.83**	**72.38**
SYSU-CD	w/o contrast	85.68	81.67	71.86	92.46	83.63
	w contrast	**82.33**	**86.09**	**72.67**	**92.69**	**84.17**

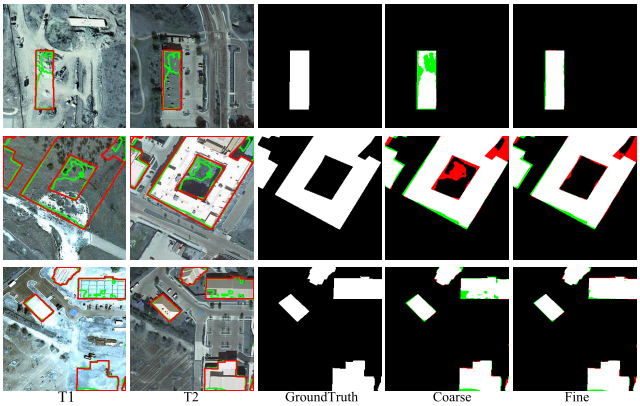


Fig. 6. Visualization of coarse-to-fine change prediction. The red lines represent the fine-level results, while the green lines indicate the coarse-level results. Red pixels correspond to FPs, and green pixels represent false negatives (FN).

TABLE VII
IMPACT OF CPFM ON MODEL PERFORMANCE

Configuration	NCA	MOE	Channel-Att	Spatial-Att	F1
1	-	-	-	-	89.74
2	✓	-	-	-	90.60
3	✓	✓	-	-	91.36
4	✓	✓	✓	-	91.55
5	✓	✓	✓	✓	**91.89**

SYSU, providing improvements in IoU by 2.58%, 1.65%, 1.38%, and 0.68%, and the corresponding enhancements in the $F1$ score by 1.11%, 0.85%, 0.74%, and 0.58%, respectively. These findings illustrate that the incorporation of coarse-level semantic cues not only increases spatial location but also enhances overall prediction accuracy, particularly in complex or ambiguous areas. In addition, the prompt-based refinement module is optional, facilitating its seamless integration into various CD frameworks without necessitating architectural alterations or imposing significant computational burdens. The visual results, as shown in Fig. 6, elucidate the efficacy of prompt guidance. Fine-grained predictions (indicated by red lines) exhibit markedly more accurate boundary demarcation than their coarse-grained counterparts (green lines), demonstrating a reduction in FPs (red pixels) and false negatives (green pixels). This visual substantiation aligns with the quantitative enhancements noted with the integration of coarse prompts. Importantly, the prompt-based refinement module is intentionally designed as an optional component, facilitating its unobtrusive incorporation into various CD frameworks. This design choice obviates the need for architectural modifications and minimizes computational overhead, rendering it a versatile and effective enhancement to existing systems.

D. Impact of HSCM

Table VI illustrates that incorporation of the HSCM produces slight quantitative enhancements, as demonstrated by an increase in IoU from 84.69% to 84.99% and an improvement in the $F1$ score from 91.71% to 91.89%. The relatively limited numerical improvement can be attributed to the dataset's dominance by easily distinguishable examples, which already achieve near-saturation performance, leaving little room for substantial metric advancement. However, the qualitative visualizations depicted in Fig. 7 elucidate the practical importance of incorporating the HSCM. Compared to the baseline approach ("without contrast"), the HSCM-enhanced model ("with contrast") proficiently attenuates activations in regions without change, while markedly amplifying responses related to areas of actual change. This capability allows for a more precise and robust differentiation between changed and unchanged features, resulting in sharper, clearer, and more discriminative feature representations. Such qualitative improvements are especially beneficial in challenging scenarios involving small-scale changes, densely distributed objects, and ambiguous boundary regions, highlighting the effectiveness of HSCM in fine-grained CD for complex remote-sensing tasks.

E. Impact of CPFM

To further examine the contribution of each submodule in CPFM, we conduct the ablation study shown in Table VII. Starting from the baseline without CPFM (Configuration 1), introducing NCA improves the $F1$ score from 89.74% to 90.60%, showing its effectiveness in bi-temporal feature alignment. Adding the MoE module further raises the $F1$ score to 91.36%, indicating its ability to adaptively fuse temporal differences. With the subsequent introduction of Channel Attention and Spatial Attention, the performance is further improved to 91.55% and 91.89%, respectively. These results demonstrate that the submodules within CPFM provide complementary gains and progressively enhance discriminative feature learning.

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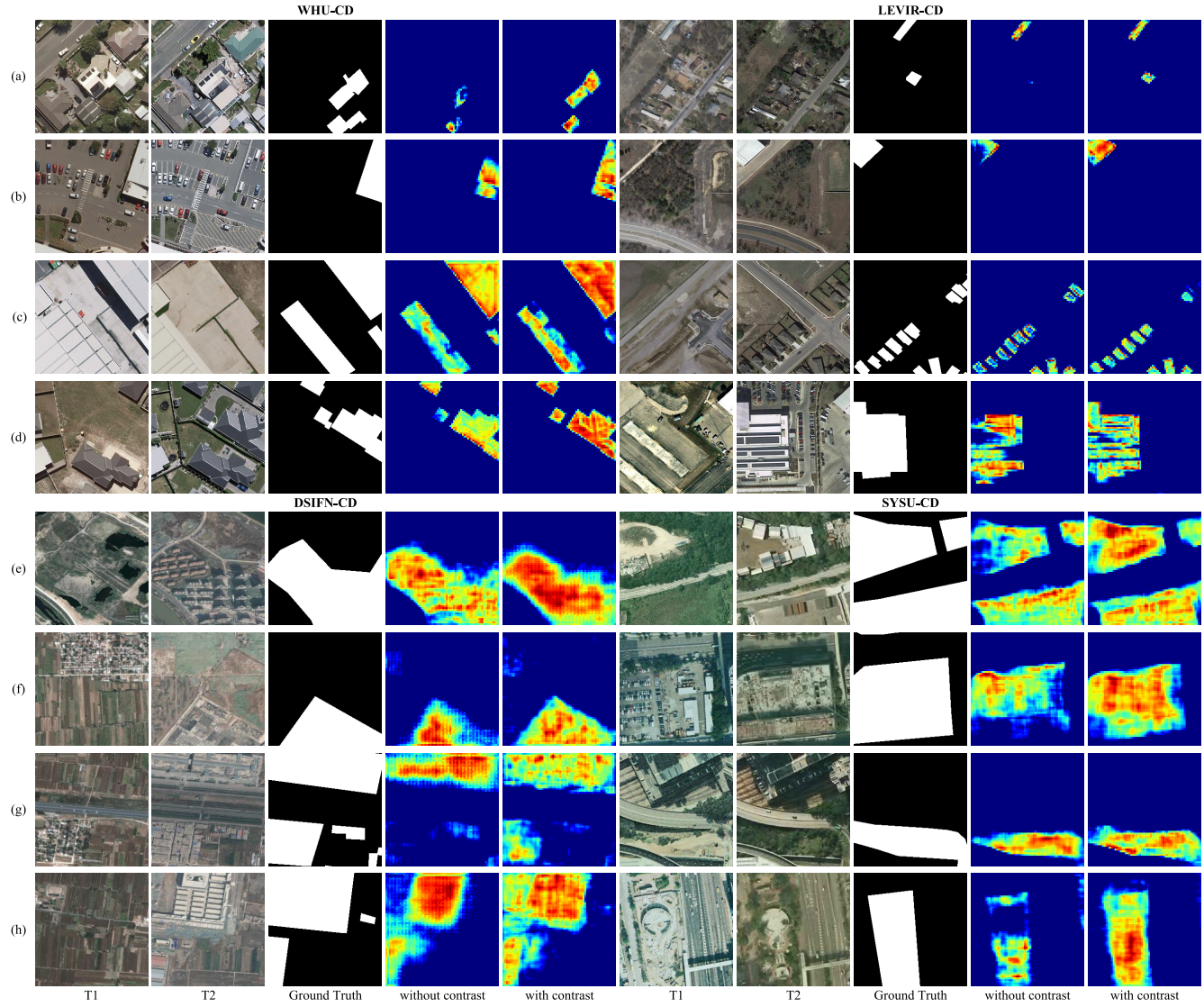


Fig. 7. Visualization of HSCM Features. (a)–(d) Show building change examples from the WHU-CD and LEVIR-CD datasets, while (e)–(h) show complex land-cover change examples from the DSIFN-CD and SYSU-CD datasets. From left to right, each group shows the T1 image, T2 image, ground truth, response map without contrast, and response map with contrast. The proposed HSCM module enhances feature discrimination by emphasizing subtle changes and reducing false activations.

TABLE VIII
ABLATION STUDIES ON DIFFERENT EXPERT COMBINATIONS

Expert x_1	Expert x_2	Expert $(x_1 - x_2)$	Expert $\lvert x_1 - x_2 \rvert$	F1
-	-	✓	-	89.22
-	-	-	✓	89.70
-	-	✓	✓	90.35
✓	✓	✓	✓	**91.89**

To further analyze the MoE design, Table VIII reports the results of different expert combinations. Using only $(x_1 - x_2)$ or $|x_1 - x_2|$ yields limited performance, with $F1$ scores of 89.22% and 89.70%, respectively. Combining both difference-based experts improves the $F1$ score to 90.35%, indicating that directional and magnitude information are complementary. When the original temporal features x_1 and x_2 are further introduced, the $F1$ score reaches 91.89%. This confirms that the proposed multiexpert design can better capture complex changes by jointly preserving temporal semantics and difference information.

VI. DISCUSSION

In this study, we investigate a structured semantic–structural fusion strategy for fine-grained CD to boost performance in regions where spectral or spatial cues are difficult to separate. Experiments on four public benchmarks confirm the effectiveness of the proposed *ChangeFine* framework. The SCFE narrows the gap between geometric structure and high-level semantics, whereas the CPFM and HSCM strengthen bi-temporal interaction and spotlight uncertain pixels for sharper boundary discrimination. The plug-and-play PCD further refines change localization by combining explicit predictions with implicit semantic differences in a coarse-to-fine manner. Compared with recent state-of-the-art models (e.g., ChangeClip and ChangeMamba), ChangeFine achieves higher quantitative scores and markedly better visual consistency in scenes containing small objects or ambiguous edges, improvements that stem from its ability to leverage object-level relations and dynamically focus on challenging regions.

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However, there are several limitations. First, the pipeline currently depends on external backbones (SAM and CLIP), which raises the computational cost. Second, the method presumes well-labeled bi-temporal imagery and reliable semantic annotations, conditions that are not always met in practice.

To address these limitations, we will investigate lightweight knowledge-distillation techniques that transfer SCFE into a compact, task-specific backbone, and we will also explore plug-in adapter modules that provide efficient and domain-adaptive fine tuning without invoking the full external models. Second, we will explore an open-vocabulary change-detection strategy that leverages large vision–language models to generate class-agnostic prompts on the fly, thereby relaxing the need for predefined labels and enabling the framework to handle unseen categories and weak or noisy annotations.

VII. Conclusion

In this article, we present a novel binary CD framework that integrates CLIP/SAM feature encoders, cross-temporal mixed expert feature fusion, hard-sample contrastive learning, and change-prompt decoding strategies to enable coarse-to-fine CD. Extensive experiments across multiple benchmark datasets demonstrate that our approach consistently outperforms state-of-the-art methods in key metrics, such as IoU and $F1$ score, highlighting its robustness and effectiveness. Looking ahead, we will compress the model and extend it to multimodal and open-vocabulary CD, further broadening its applicability and impact on intelligent remote-sensing systems.

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